

Final Project Plan / Ethics Review

I'm basing my final project on the Wang, et al. paper "Structure of 311 service requests as a signature of urban location."¹ This paper asks if 311 service request patterns predict socioeconomic characteristics of neighborhoods in large urban cities, and specifically looked at whether neighborhoods with similar demographics (like income, race, education) produce similar 311 complaint patterns. The paper also aimed to learn if these requests could be used to create an accurate model of socioeconomic performance of urban neighborhoods in general, as a tool for "urban stakeholders to quantify the impacts of their interventions." (1)

Wang, et al. did a large-scale analysis of 311 non-emergency service request data from New York, Chicago, and Boston from 2012-2015. They used machine learning models to group census tracts by relative frequencies of complaint types, and created "complaint signatures" from this grouping. They then compared the data against large demographic data from the American Community Survey (ACS) census. They found that neighborhoods in these cities do have distinct signatures—wealthier areas typically report things like noise complaints, and lower-income areas have higher frequencies of blocked driveway and heating issue complaints—and that 311 data patterns could very accurately predict factors like median income, education level, and the racial demographics of an area as they're represented in ACS census data.

I'm going to be recreating parts of this research, but for 311 data in the neighborhoods of Pittsburgh, Pennsylvania. I'm going to compare Pittsburgh 311 requests with ACS census data to see if correlations appear based on socioeconomic characteristics of Pittsburgh neighborhoods. We've seen that 311 data can be effective in extremely densely populated, metropolitan areas like New York (population: ~8.5m), but can 311 service request patterns in smaller cities like Pittsburgh (population: ~300k) predict socioeconomic characteristics of its neighborhoods in a similar way the data does in much larger cities?

For this research, I'll need public 311 service request data from Pittsburgh and ACS census data with the census tracts of neighborhoods in Pittsburgh. To obtain this data, I'll be using the Western Pennsylvania Regional Data Center's (WPRDC) public 311 data archive on [wprdc.org](https://data.wprdc.org)² and the United States Census Bureau's online data table tool on [census.gov](https://data.census.gov),³ and

¹ Wang, et al. 2017. "Structure of 311 Service Requests as a Signature of Urban Location."
<https://doi.org/10.1371/journal.pone.0186314>.

² WPRDC 311 Data: <https://data.wprdc.org/dataset/311-data/resource/29462525-62a6-45bf-9b5e-ad2e1c06348d>

³ United States Census Bureau online data table:
[https://data.census.gov/table?q=DP03&q=160XX00US4261000,4261000\\$1400000&d=ACS+5-Year+Estimates+Data+Profiles](https://data.census.gov/table?q=DP03&q=160XX00US4261000,4261000$1400000&d=ACS+5-Year+Estimates+Data+Profiles)

will grab ACS 5-year estimate data for the census tracts within Pittsburgh, PA. Additionally, since the ACS data does not provide census tract details in the same way that the WPRDC 311 data does, I'll probably need some way of cleaning and "converting" this data so I can actually compare it. Luckily, a list of neighborhood names to census tracts already exists; it was done by the Government Information Librarian, Christopher Lemery, at the University of Pittsburgh.⁴ I'm not exactly sure how I'll go about this process yet, but this is a good start.

This project is worthwhile because 311 and other similar request data is such a helpful big data source. It's public, it's huge (depending on the place, but in general there are many data points), and it's pretty granular too. If we can draw correlations between this data and census data to accurately represent the socioeconomic characteristics of neighborhoods in cities like Pittsburgh, it has the potential to be a really useful tool for a lot of research and business. 311 data doesn't replace census data or other data sources, but it's much more timely than census data and free to use to inform analysis. It's a really helpful tool! Being able to verify its validity might be a very big help for urban planners and researchers looking to use this data to inform business decisions and research. The original paper focused on the biggest cities in the U.S., and my research will scale this down to see if we can generalize the use of 311 data to smaller but still urban areas like Pittsburgh, since we don't know if people in areas this size use 311 data like people do in NYC, for example.

I anticipate running into challenges figuring out exactly how to map census tracts to the correct neighborhoods. I also anticipate challenges figuring out the format of both datasets in general, since I've never used either before. I already ran into other challenges in the process of creating this initial plan. I tested much of this initial plan out on Minneapolis, MN, but I realized later in the process that their 311 data source didn't have neighborhood names in the data, which pretty much invalidated that source for my use case. I also tried Detroit and Ann Arbor, but the data wasn't granular enough, and/or the data wouldn't be very easy to access or work with. I was going to be having to do a ton of API calls with Ann Arbor through an API I wasn't familiar with at all, for example, for a project that would have been very limited by the small scope of the data. If my data access fails and it doesn't work with Pittsburgh, I'll go back and search for a different city's 311 data, and repeat with that city. Now that I know how to use the census.gov's data table tool and how to find 311 data for other cities, that would be a relatively simple change. If I can't find any cities similar in size to Pittsburgh that work, I'll go back to a large city like Chicago or Boston and perform my analysis on those cities, and try to see if there are other correlating socioeconomic factors that 311 patterns can predict. In terms of my timeline for data collection, I've already done it. I already downloaded the 311 data from census.gov and wprdc.org.

As proof of being able to access this data, here are screenshots of the WPRDC 311 data and the ACS census data:

⁴ Neighborhood names to census tracts list: <https://pitt.libguides.com/pghcensus/pghcensustracts>

311 data:

The image shows the Western Piedmont Regional Data Center's 311 Data Portal on the left and an Excel spreadsheet of the data on the right. The portal displays a table with columns for ID, Group ID, Request count, Parent closed, Status, Status code, and Department. The Excel spreadsheet has columns A through K, with headers for ID, Group ID, Request count, Parent closed, Status, Status code, Department, Request type, Request date, and Request status. The data in the spreadsheet is a direct export of the data shown in the portal's table.

ID	Group ID	Request count	Parent closed	Status	Status code	Department	Request type	Request date	Request status
1	827262	1	False	open	0	DWP - Administration	311	2024-03-12	Open
4	846913	1	False	open	0	Police - Zone 1-6	311	2024-03-12	Open
5	853798	1	False	open	0	Police - Facilities	311	2024-03-12	Open
6	882180	1	False	open	0	311	311	2024-03-12	Open
12	873927	1	False	In progress	3	DWP - Forestry Division	311	2024-03-12	In Progress
19	868766	1	False	In progress	3	DWP - Refuse	311	2024-03-12	In Progress
21	874963	1	False	open	0	DWP - Administration	311	2024-03-12	Open
25	826464	1	False	open	0	DWP - Administration	311	2024-03-12	Open
29	833900	1	False	open	0	DWP - Support Services	311	2024-03-12	Open
35	872006	1	False	open	0	DWP - Refuse	311	2024-03-12	Open

ACS data:

The image shows the US Census Bureau's DP03 Selected Economic Characteristics table for Census Tract 103.01, Allegheny County, Pennsylvania, on the left and an Excel spreadsheet of the data on the right. The table displays various economic characteristics such as Population, Labor force, and Unemployment rate. The Excel spreadsheet has columns A through K, with headers for Label, Total, and various economic characteristics. The data in the spreadsheet is a direct export of the data shown in the table.

Label	Total	Unemployment rate	Population	Labor force	Unemployed	Armed Forces	Not in labor force	Civilian labor force	Unemployment rate
Population	1,855 ±55		1,855 (X)						
Labor force	0 ±12		0 (X)						
Civilian labor force	0 ±12		0 (X)						
Unemployed	0 ±12		0 (X)						
Armed Forces	0 ±12		0 (X)						
Not in labor force	1,855 ±55		1,855 (X)						
Civilian labor force	0 ±12		0 (X)						
Unemployment rate	0 (X)	0 (X)							

There are absolutely ethical tensions involved with using 311 data. However, I've considered Salganik's four ethical principles when planning this research. In terms of Respect for persons, the data I'm using is de-identified and public. However, reidentification risk does exist if granular location data are linked with census tract data. Beneficence is high in this research. 311 is a free data source and tool that can be used for urban planning and by researchers. Justice is a point of tension, though. If 311 data is used to decide where to allocate resources, neighborhoods that don't have access to tools needed to submit 311 requests (or that don't know about or know how to use these tools) will be less represented in this data. In other words, already-marginalized groups might be made more marginalized in some aspects if

relying on this data. Finally, respect for law and public interest is upheld by using publicly available data on open-data portals and public US government census sites. I'm using this data according to the terms of service of this site and mitigating ethical/privacy concerns that might show up while doing this type of research. This project has pretty similar ethical implications to the rest of the paper. I'm taking publicly available 311 data and ACS census data and finding patterns in it, and seeing if correlations show up, so nothing has really changed there.

The main benefactors of this research would be researchers looking to use 311 data in smaller yet still urban areas like Pittsburgh, and any urban stakeholders or planners. Citizens would in turn benefit from this, but this research doesn't directly benefit them. This research centers the voices of more "active" citizens with the ability and resources to report issues through this 311 channel, and marginalizes communities with less access to resources like electronics and internet connections, as well as those with lower trust in the government who might not be compelled to report issues to government channels. A potential harm / unintended consequence might be development companies using this data to unfairly categorize neighborhoods. This sort of "redlining" might be used to discriminate against minority populations by these types of institutions.

In order to mitigate any privacy issues associated with the use of 311 data, I will make sure to not grab unnecessary data like the street the request was submitted on, or specific longitude/latitude coordinates that aren't needed for my analysis. I will not use any specific addresses or timestamps that could track individuals or their routines (like commutes, etc.) either. Additionally, I will also include a section in my report talking about the limits of 311 data, like I've covered in the above sections.